

Authentizotic climates and employee happiness: Pathways to individual performance?

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Abstract

This article discusses how six authentizotic psychological climates explain stress and affective well being at work, and how stress and affective well being explain self-reported individual performance. The sample comprises 199 employees from 118 organizations. The main findings include the following points. The psychological climates explain unique variance of stress, affective well being and performance. Stress explains unique variance of affective well being. Affective well being, mainly enthusiasm, vigor and placidity explain unique variance of performance.

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1. Introduction

Gavin and Mason (2004) observe that economic productivity is often extracted from the average worker, in large measure, at the cost of his/her health and happiness and such dysfunctional effects needs to end. The “positive organizational studies” movement (Cameron, Dutton, and Quinn, 2003) suggests that more efforts must be devoted to developing people’s strengths, to help people achieve happiness and to contribute to making life meaningful. Gavin and Mason (2004) illustrate their arguments with companies ranked in Fortune’s “100 best companies to work for”. This ranking considers credibility, respect, fairness, pride and camaraderie as the main characteristics of the good workplace. The ranking affords the idea a great deal of attention, with featured companies serving as potential benchmarks to organizations in search of improvement and good publicity. The

presence in the ranking may be a significant source of reputation in the job market in this era of “war for talent” (Martin et al., 2005). The model represents the popular expression of a focus on the human side of the organization that is also present in projects such as the *Best Companies for Working Mothers* and APA’s *Psychologically Healthy Workplace Award*.

The interest academia shows in the positive impact of “best places to work for” on organizational performance provides a useful dialogue between the managerial and scholarly fields (Filbeck and Preece, 2003; Fulmer, Gerhart, and Scott 2003). One prominent example of this dialogue arose with Kets de Vries’ (2001) notion of the “authentizotic organization,” to capture the essence of these workplaces. “Authentizotic” is a neologism that combines the Greek words “authentekos” and “zoteekos”, meaning that an organization is trustful and reliant, and vital to life, respectively. The idea underpinning the authentizotic theory is that organizations can be a source of meaning and growth for people, instead of leading to psychological suffering and feelings of alienation (Frost, 2003).

Organizations with strong authentizotic features (as is allegedly the case of “the best companies to work for”) are able to develop a set of meta-values that allow their members to nourish their senses of purpose, self-determination, impact, competence, belonging, meaning and enjoyment. According to

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Kets de Vries, they can be an anchor for health and psychological well being (PWB), a way of developing a positive self-esteem, and a source of coping with stress. Thus, the search for workplaces with strong authenticzotic features promises to counterbalance the bias toward the negative side of organizational life.

Empirical research on authenticzotic organizations is scarce. For example, the psychometric properties of the instrument used by The Great Place to Work Institute to rank companies are unknown outside of the Institute itself. Taking inspiration from the authenticzotic concept, Rego (2004) and Rego and Souto (2004, 2005) develop and validate an instrument for measuring authenticzotic psychological climates. They do not measure real workplace characteristics (as is usually addressed through the direct observation and/or the aggregation of individual perceptions at the organizational level of analysis), but rather employees' perceptions of those characteristics. Factor analyses suggest a well-adjusted six-factor model: spirit of camaraderie, trust/credibility of the leader, open and frank communication with the leader, opportunities for learning and personal development, fairness and work–family conciliation (see Table 2). The six climates predict turnover intentions, commitment and self-reported individual performance.

This paper shows how these six psychological climates explain two constructs related to PWB (affective well being at work and stress), and how stress predicts affective well being. As a secondary aim, self-reported individual performance is related to psychological climates and PWB. The aim is “secondary” because self-reported individual performance is susceptible to several biases, discussed later in the paper. Fig. 1 depicts the hypothesized model: (1) people who perceive their organizations as more authenticzotic experience greater affective well being and lower stress; (2) lower stress associates with greater affective well being; (3) the psychological climates predict individual self-reported performance, either through the mediation of both PWB aspects or through other aspects not studied here.

The study focuses on psychological climates, not on organizational climates (normally measured through the aggregation, at the organizational level, of individual perceptions). Psychological climates are the “individual's psychologically meaningful representations of proximal organizational structures, processes, and events” (Parker et al., 2003, p. 390). The construct of psychological climate is alive and well (Parker et al., 2003) and influences important individual-level outcomes (e.g., motivation, commitment, satisfaction and performance). Psychological climate enables people to interpret events, predict possible outcomes, and gauge the appropriateness of their subsequent actions. Variation in these perceptions and valuations is likely to result from differences among people, from differences in “real” situations, from the employee–situation interaction, and from perceptual biases (Brown and Leigh, 1996). Although employees in the same organization may share the perceptions of the work environment, different employees can also espouse different perceptions and, accordingly, react differently to the same “real” environment. Studying psychological climate seems an appropriate way to research well being because, as Haller and Hadler (2006) point out, people's subjective perception and evaluation

are most significant for happiness and satisfaction, not so much the objective situation itself.

This article has the following structure. Sections 2, 3 and 4 theoretically integrate the concepts of authenticzotic psychological climates, stress and affective well being, and discuss the relationships between PWB and self-reported individual performance. Section 5 presents the method, emphasizing the procedures for developing and validating the instrument to measure psychological climates. Section 6 presents the results and Section 7 discusses the main findings, the implications for management, the limitations of the research and some avenues for future studies. Given the negative bias of organization and management studies, the research contributes to the “positive turn” in organization and management research (Cameron et al., 2003).

2. Psychological climates and affective well being

2.1. Happiness in the workplace

Happiness is a subjective experience: people are happy to the extent that they believe themselves to be happy. Scholars tend to treat happiness as PWB, a three dimensional construct that includes life satisfaction, the presence of positive emotional experiences and absence of negative emotional experiences (Diener, 2000). Although the pathways to happiness vary across cultures (Haller and Hadler, 2006), the quest for happiness is universal. Scholars such as Diener and Seligman (2004) and Kahneman et al. (2004) advocate the creation of national well being accounts to complement national income accounts. Workplaces play a crucial role in people's happiness, as Gavin and Mason (2004, p. 381) point out: “It seems clear that if there is any hope for people to find general happiness in their lives today, they must be happy at work. Work by itself, of course, cannot make a person happy, but a person cannot be genuinely happy if he or she is unhappy at work”. Therefore, promoting employees' PWB is intrinsically good (e.g., Cameron et al., 2003) and also a good way to promote individual and organizational performance (Daniels and Harris, 2000; Wright and Cropanzano, 2000, 2004; Wright, Cropanzano, Denney, and Moline, 2002).

Such likely consequences of employees' PWB are stimulus for researchers to study its antecedents. This paper focus on affective well being, one of the most important indicators of PWB (Daniels, 2000). The construct reflects the frequent experience of positive affects and the infrequent experience of negative affects, comprising five bi-polar dimensions: anxiety–comfort, depression–pleasure, boredom–enthusiasm, tiredness–vigor and anger–placidity. As discussed next, the literature suggests that the perceptions of authenticzotic climates may influence affective well being.

2.2. Spirit of camaraderie

Gratifying relationships with other people are a major source of PWB (Baumeister and Leary, 1995; Haller and Hadler, 2006). They play a crucial role in meeting social, intimacy and

security needs. They are important sources of social support, which relates to different measures of PWB, such as positive and negative affect, self-esteem, adjustment, comfort, enthusiasm, loneliness, job satisfaction, health, neuroticism and happiness (Christopher, Kuo, Abraham, Noel, and Linz 2004; Daniels, 2000; Kiefer, 2005; Limbert, 2004). Thus, positive perceptions of spirit of camaraderie tend to make employees feel that they can meet important affiliation and social needs, experiencing greater affective well being. Hence: H_{1a}. Employees with positive versus negative perceptions of spirit of camaraderie experience greater affective well being.

2.3. Positive leader-follower relationships

Trust and credibility of the leaders, as well as open and frank communication with them, may strengthen the employees' feelings of emotional support, improving their well being (Aycan and Eskin, 2005; Kramer and Tyler, 1996). They can also promote employees' self-esteem and identification with the organization, leading to more pleasant affects (Herrbach and Mignonac, 2004). On the contrary, when trust and communication with the leader are poor, employees may feel pressed to express emotions that differ from those that they privately feel, thus experiencing emotional dissonance and lower PWB (Morris and Feldman, 1996). The perceptions of trustful behaviors of leaders also favor more cooperative behavior among colleagues, which can further lead to pleasant affects (Herrbach and Mignonac, 2004). Employees also likely consider their leaders as authentic thus identifying more strongly with them and with the organization and experience more positive emotional states and greater self-realization (Ilies, Morgeson, and Nahrgang, 2005). From this, the next hypotheses follow. H_{1b}. Employees who perceive that leaders are high versus low in being trustworthy and credible experience greater affective well being. H_{1c}. Employees who perceive that they can communicate openly and frankly with leaders experience greater affective well being than those employees who do not have such perceptions.

2.4. Opportunities for learning and personal development

Positive perceptions of opportunities for learning and personal development may render the job more intrinsically rewarding and, thus, lead to feelings of well being (Diener and Suh, 1999; Kasser and Ryan, 1996). Such perceptions may also lead to an increase in the perceived meaningfulness of work, encouraging people to invest more cognitive and emotional resources in their work, and enhancing employee identification with their work roles and organizations (Brown and Leigh, 1996). These effects, in turn, can lead to positive affects (Herrbach and Mignonac, 2004). People also develop stronger senses of job competence and autonomy when they perceive learning opportunities, inducing them to feel more enthusiastic and comfortable in the presence of the job requirements (Daniels, 2000). H_{1d}. Employees who perceive good versus poor opportunities for learning and personal development experience greater affective well being.

2.5. Justice

Justice perceptions likely associate with affective well being (Brief and Weiss, 2002). For example, Weiss et al. (1999) argue that justice perceptions influence happiness, guilt, anger and pride. Fitness (2000) suggests that followers' anger is a product of unjust treatment by leaders. Tepper (2001) points out that perceptions of justice predict depression and emotional exhaustion. According to this author, perceptions of injustice may threaten the employees' senses of respect and dignity and their well being, harming their self-worth and self-efficacy, leading them to feel a lack of the psychological coping resources needed to sustain hope and problem solving in the face of threats to their well being. The consequence is that employees will regard injustice factors as stressors that produce psychological distress. H_{1e}. Employees with positive versus negative justice perceptions experience greater affective well being.

2.6. Work-family conciliation

Work-family conflict decreases career and life satisfaction, impoverishes marital adjustment, and increases unhappiness, stress, depression, anxiety and substance abuse (Aycan and Eskin, 2005; Greenhaus, Allen, and Spector, 2006; O'Driscoll, Brough, and Kalliath, 2004). Therefore, the perceptions of lack of work-family conciliation can lead to poorer quality of life and lower levels of affective well being. Such perceptions may induce employees to feel a lack of organizational support, generating lower affective well being (Deborah, Michelle, and Linda, 1993; Richardsen, Burke, and Mikkelsen, 1999). Employees can also lose the sense of meaningfulness at work (Richardsen et al., 1999). On the contrary, when they perceive conciliation between both roles, they engage more strongly in work and family roles, meet their needs in both of them, experience less stress when participating in both roles, and obtain high self-esteem from the competence they achieve in their family and working lives (Marks and MacDermid, 1996). H_{1f}. Employees with positive versus negative perceptions about work-family conciliation experience greater affective well being.

3. Antecedents and consequences of stress

Stress is the inability to cope with one's job pressure (Ganster and Schaubroeck, 1991). Stress emerges, for example, when an individual (a) perceives that external demands are not matched by internal needs and values, (b) senses a lack of psychological resources to face those demands or enough discretion to control the job, or (c) imputes high importance to the perceived demands and considers that insufficient social support to meet them exists. Work stress is also likely associated with low levels of job satisfaction and PWB (Daniels and Guppy, 1994; Deborah et al., 1993; Fairbrother and Warn, 2003). H₂. Employees with higher versus lower stress experience lower affective well being.

Several aspects of working life relate with stress (Fairbrother and Warn, 2003; Ganster and Murphy, 2000; Greenberg, 2004; Hart and Cooper, 2001; Spector, Cooper, Poelmans, and 12 other authors, 2004): unsatisfactory interpersonal relationships,

autocratic leadership behavior, work–family conflict, absence of social support, lack of involvement in decision making, lack of opportunities for career advancement, lack of opportunity to use personal talent, lack of control over aspects of the job, and unfairness. However, as the transactional models suggests, individuals perceive and deal differently with the same stimuli. Employees experience greater stress when they perceive poor authentic climates. Such perceptions may decrease the employee's sense of social support (Daniels and Guppy, 1994; Richardsen et al., 1999), an important asset for individual adjustment, which relates to superior health and well being (Aycan and Eskin, 2005; Nielson, Carlson, and Lankau, 2001).

Perceptions of poor spirit of camaraderie may also make people feel that external constraints prevent them from meeting social needs, and that they have insufficient coworker support to deal with difficult or challenging tasks. The perceptions of leaders' disrespectful behaviors may make employees feel a lack of social support for coping with job demands, fear of being betrayed, and lack of voice opportunities. The perceptions of lack of opportunities for learning and personal development may signal poor job autonomy, lack of control over aspects of the job (Bussing, Bissels, Fuchs, and Perrar, 1999; Fairbrother and Warn, 2003) and lack of opportunities for career advancement. Perceptions of unfairness may lead to feelings of poor relationships with leaders, lack of psychological safety, a feeling of being disrespected, and feelings of lack of social support for coping with the events that have implications for employee well being (Greenberg, 2004; Tepper, 2001). The perceptions of lack of work–family conciliation may give rise to feelings of role conflict, and to the sense that the external demands are outside employee control. On the contrary, positive perceptions of authentic climates may facilitate the employees' strengths of resilience and hope (Norman, Luthans, and Luthans, 2005), in turn providing them with resources to deal more proactively and positively with stressful demands and challenges. The following set of hypotheses follows from this discussion. H_{3a}. Employees with positive versus negative perceptions of spirit of camaraderie experience lower stress. H_{3b}. Employees with positive versus negative perceptions of trust and credibility of the leader experience lower stress. H_{3c}. Employees with positive versus negative perceptions of open and frank communication with the leader experience lower stress. H_{3d}. Employees with positive versus negative perceptions of opportunities for learning and personal development experience lower stress. H_{3e}. Employees with positive versus negative perceptions of fairness/equity experience lower stress. H_{3f}. Employees with positive versus negative perceptions of work–family conciliation experience lower stress.

4. Psychological climates, happiness and performance

4.1. The value of self-reported individual performance

People are prone to hold favorable views of their own performance (Fox and Dinur, 1988; van der Heidjen and Nijhof, 2004). When compared to objective criterion measures, self-ratings tend to be less accurate than ratings from peers and supervisors (van der Heidjen and Nijhof, 2004). However, self-

assessments can be of great value, namely when anonymity is guaranteed and/or individuals perceive no need to present themselves favorably for career, performance appraisal and/or social acceptance purposes (van der Heidjen and Nijhof, 2004). Fox and Dinur (1988) argue that individuals possess the ability to evaluate themselves reliably in a manner similar to that of others and in a way that can predict subsequent performance. Van der Heidjen and Nijhof (2004) argue “despite doubts on the validity of self-assessment and its utilization on applied research settings, there is reason to believe that individuals are in a good position to make a valid assessment of their own knowledge and capabilities” (p. 495). Levine (1980) states “we all have a good deal of time to verify, logically and rationally, our feelings about ourselves and our performance, because we have lived with ourselves since we can remember” (p. 261). The author argues: (a) individuals experience the effects of their actions directly in a number of real-world settings; (b) they gain information about things in the world around them by observing others; (c) they also receive others' judgments of their performance; (d) they have a good deal of time to verify feelings about themselves and their performance.

Thus, on the whole, employees who self-describe as more/less productive are likely described by their supervisors as more/less productive as well and are “really” more/less productive according to objective standards. The method limitations deriving from this choice can make findings less robust.

4.2. Happiness as a pathway for performance

Promoting employees' PWB can be a good way of stimulating individual and organizational performance. Several positive emotions, including the experience of PWB, can broaden the scope of attention, cognition, and action, and build physical, intellectual and social resources (Fredrickson, 2001). Positive emotions can also foster employees' perceptions of enhanced meaning from work (Wright and Cropanzano, 2004), inducing them to work not just for financial rewards or career advancement, but also for the personal fulfillment of “doing a good job”. And this orientation can promote their performance. As Gavin and Mason (2004) argue: (1) when a workplace creates meaning for workers, they tend to be more healthy and happy; (2) being healthy and happy, employees tend to be more productive over the long run; (3) health, happiness and productivity are essential ingredients of a good society; (4) the search for productivity alone, which is almost the sole emphasis of many organizations, is not enough.

Positive emotions can also help people to thrive when confronted with adverse situations, to be more proactive and resilient, less prone to stress symptoms and more disposed to develop productive social connections. As a consequence, the work climate improves and productivity increases. When individuals experience positive affect, they are more eager to help others who are in need (Fredrickson, 1998; Isen, 1987), stimulating the positive emotion of gratitude in the people who receive help. These experiences of gratitude, in turn, often create the urge to reciprocate, thus amplifying the effects of the first helping act and reinforcing the possibilities of higher

performance (Cameron, Bright, and Caza, 2004). H₄. Employees with higher versus lower psychological well being report higher individual performance.

4.3. Stress as predictor of performance

Stress can influence performance through the mediating role of happiness (i.e., low stress associates with higher PWB, and higher PWB associates with better performance). However, stress can relate with individual performance without the mediating role of PWB. For example, greater stress can impoverish the cognitive processing of information and lead to worse decisions. Stress may have a detrimental impact upon employees and the organization (Cooper, 1998; Evers, Frese, and Cooper, 2000; Ganster and Murphy, 2000; Hart and Cooper, 2001; Lambert, Lambert, and Yamase, 2003; Stall, 2004). Consequences include absenteeism, wasted time, sabotage, theft and damage, performance decline, accidents at work, higher costs with healthcare, low motivation levels, increased interpersonal conflicts, communication failures and decision making mistakes. H₅. Employees with higher versus lower stress report lower individual performance.

4.4. Psychological climates as predictors of individual performance

Psychological climates may affect individual performance directly or through other mediators not studied here. For example, perceptions of open and frank communication with the leaders may improve the quality of leader–member exchange (Mueller and Lee, 2002; Yrle, Hartman, and Galle, 2003) and promote individual performance (Dunegan, Uhl-Bien, and Duchon, 2002; Michael, Harris, Giles, and Feild, 2005). The perceptions of opportunities for learning and personal development may induce employees to obtain meaning from performing the job, to feel that they can develop their competencies, and to put their full potential into carrying out the job. This view is consistent with a prominent stream of literature arguing that employees engage more completely and invest their entire selves in the job when they perceive that the work environment is psychologically safe and meaningful and provides conditions to satisfy their psychological needs (Brown and Leigh, 1996). Martin et al. (2005, p. 263) also observe that “employees whose perceptions of the organization and environment in which they were working (...) were more positive, were more likely to appraise change favorably and report better adjustment in terms of higher job satisfaction, psychological well being, and organizational commitment, and lower absenteeism and turnover intentions”. Parker et al. (2003) also suggest that psychological climates “do have reliable relationships with employees’ work attitudes, psychological well being, motivation and performance” (p. 405). The following set of hypotheses follows from this discussion. H_{6a}. Employees with positive versus negative perceptions of spirit of camaraderie report greater individual performance. H_{6b}. Employees with positive versus negative perceptions of trust and credibility of the leader report greater individual performance. H_{6c}. Employees with positive versus negative perceptions of open and frank

communication with the leader report greater individual performance. H_{6d}. Employees with positive versus negative perceptions of opportunities for learning and personal development report greater individual performance. H_{6e}. Employees with positive versus negative perceptions of fairness/equity report greater individual performance. H_{6f}. Employees with positive versus negative perceptions of work–family conciliation report greater individual performance.

5. Method

5.1. Sample and procedures

The study uses a convenience sample of 213 employees from 128 organizations. The individuals are participants of a seminar on ethics and organizational behavior, having participated in the study before the seminar. They are all university graduates and perform a wide range of jobs (e.g., bank and insurance clerks,

Table 1
Confirmatory factor analysis: affective well being*

	First-order factor model	Second-order factor model
Anxiety–comfort	(0.71)	(0.71)
Anxious (r)	0.69	0.72
Worried (r)	0.55	0.56
Tense (r)	0.74	0.70
Depression–pleasure	(0.75)	(0.75)
Depressed (r)	0.68	0.65
Happy	0.85	0.87
Cheerful	0.65	0.66
Boredom–enthusiasm	(0.89)	(0.89)
Enthusiastic	0.86	0.84
Motivated	0.86	0.88
Optimistic	0.83	0.84
Tiredness–vigor	(0.72)	(0.72)
Active	0.92	0.90
Alert	0.46	0.46
Full of energy	0.74	0.76
Anger–placidity	(0.72)	(0.72)
Aggressive (r)	0.51	0.49
Calm	0.73	0.70
Annoyed (r)	0.83	0.87
Overall affective well being		(0.85)
Anxiety–comfort		0.40
Depression–pleasure		0.90
Boredom–enthusiasm		0.95
Tiredness–vigor		0.77
Anger–placidity		0.61
Fit indices		
Chi-square/degrees of freedom	2.1	2.0
Root mean square error of approximation	0.07	0.09
Goodness of fit index	0.90	0.86
Adjusted goodness of fit index	0.85	0.81
Comparative fit index	0.93	0.89
Incremental fit index	0.93	0.89
Relative fit index	0.85	0.80

*Completely standardized solution.

(r) Reverse-coded items.

In brackets: Cronbach alphas.

engineers working in several construction and telecommunications companies, computer programmers and operators, salespeople). Due to non-response, 14 individuals are not considered. The final sample includes 199 employees (28% female) from 118 organizations. Mean age is 33.4 years, and tenure is 7.6 years. To reduce common method biases, respondents' answers were anonymous and individuals were assured that no right or wrong answers existed. Counterbalancing the order of the measurement of the predictor and criterion variables was also performed. Different scale endpoints, formats and ranges for the predictor and criterion measures were employed.

5.2. Self-reported individual performance, stress and affective well being measures

First, self-reported individual performance is measured with three items from Staples et al. (1999), and one worded by us. The items are: (1) "I believe I am an effective employee"; (2) "I am happy with the quality of my work output"; (3) "My manager believes I am an efficient worker"; (4) "My colleagues believe I am a very productive employee". Individuals report the degree to which each assertion apply to them, on a seven-point Likert scale (1 = does not apply to me at all; 7 = applies completely to me). Cronbach alpha is 0.86.

Second, stress is measured with three items from Staples et al. (1999), and eight from the authors. Sample items are "I work under a lot of tension" and "Problems associated with my job have kept me awake at night". Individuals report whether statements apply to them, on a seven-point scale. Cronbach alpha is 0.85.

The study measures affective well being with the instrument validated by Daniels (2000). The instrument includes 30 bi-polar scales, measuring five dimensions: anxiety–comfort, depression–pleasure, boredom–enthusiasm, tiredness–vigor and anger–placidity. Participants were invited to think about their feelings over the last three months in the organization, and to answer on a seven-point scale ranging from "never" (1) to "always" (7). Confirmatory factor analysis shows that the five-factor model fits the data unsatisfactorily. After removing some items according to the modification indices and standardized residuals, a well-fitted 15-item model emerged (Table 1). All alphas are greater than 0.70. For parsimony, a second-order factor analysis was also carried out, with the five well being factors as measuring indicators. The model fits the data moderately well, and Cronbach alpha of the second-order factor (overall affective well being) is 0.85.

5.3. Psychological climates scale development

Finally, participants report their psychological climates with an instrument previously developed and validated by Rego and Souto (Rego, 2004; Rego and Souto, 2004, 2005). Scale development started with wording and collecting items on several positive reports in the literature, including "authentic organizations" (Kets de Vries, 2001), "best places to work for" (Levering and Moskowitz, 1993), "psychological sense of community at work" (Burroughs and Eby, 1998), "living companies" (De Geus, 1997), and "family-friendly companies" (e.g., Strachan and Burgess, 1998). Items were developed to

meet the seven senses that, according to Kets de Vries, are essential for creating workplaces with authentic features and for promoting employee health: purpose, self-determination, impact, competence, belonging, meaning and enjoyment. These senses satisfy two motivational need systems of individuals that are of particular interest for life in organizations: attachment/affiliation, and exploration and assertion (Kets de Vries, 2001).

After exploratory and confirmatory analyses, an instrument comprising 21 self-report scales and measuring six authentic dimensions (Table 2) emerged. These dimensions fit the seven senses referred to by Kets de Vries. The spirit of camaraderie fits the sense of belonging. Equity, being focused on fair treatment and outcome distribution, fits the senses of belonging (as suggested by the group-value model; Tyler, 1989; see also de Cremer and Blader, 2006, about the relationships between justice

Table 2
Psychological authentic climates: confirmatory factor analysis*

Spirit of camaraderie	(0.77)
A sense of family exists among the employees.	0.61
People show concerns for the well being of the others.	0.58
A great team spirit characterizes the organization.	0.80
The organization atmosphere is friendly.	0.75
Trust and credibility of the leaders	(0.80)
People trust in their leaders.	0.75
Leaders fulfill their promises.	0.74
People feel that the leaders are honest.	0.76
Open and frank communication with the leader	(0.76)
People feel free to communicate frankly and openly with the leaders.	0.74
People feel free to show discordances to their leaders.	0.85
Talking with people placed at higher positions in the organization is easy.	0.62
Opportunities for learning and personal development	(0.77)
People feel that they can learn continuously.	0.61
People can place their creativity and imagination in benefit of the work and the organization.	0.61
People feel that important responsibilities are assigned to them.	0.59
People feel that they can develop their potential.	0.89
Fairness/justice	(0.69)
When good outcomes are reached through the employee's efforts, the "laurels" (e.g., compensation and praise) are distributed only to a few managers.(r)	0.65
People feel discriminated.(r)	0.69
Personal favoritism in the promotions exists. (r)	0.63
Work–family conciliation	(0.84)
This organization helps employees to reconcile work and family life.	0.84
The organization acts in order to allow people to conciliate work with their family responsibilities.	0.89
For advancing in the career, one needs to sacrifice family life. (r)	–0.49
The organization creates conditions so that people can maintain their children's instruction.	0.80

Fit indices

Chi-square/degrees of freedom	2.1
Root mean square error of approximation	0.07
Goodness of fit index	0.86
Adjusted goodness of fit index	0.81
Comparative fit index	0.91
Incremental fit index	0.91
Relative fit index	0.81

*Completely standardized solution.

(r) Reverse-coded items.

In brackets: Cronbach alphas.

Table 3
Means, standard deviations and correlations

	Mean	SD	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Spirit of camaraderie	4.1	0.8	–												
2. Trust and credibility of the leader	4.2	1.0	0.65	–											
3. Open and frank com. with the leader	4.2	1.0	0.57	0.67	–										
4. Op. for learn. and pers. development	4.3	0.8	0.53	0.62	0.56	–									
5. Fairness	4.8	1.0	0.47	0.58	0.62	0.46	–								
6. Work–family conciliation	3.8	1.1	0.37	0.50	0.36	0.25	0.30	–							
7. Stress	2.8	0.8	–0.39	–0.48	–0.29	–0.21*	–0.24	–0.41	–						
8. Comfort	3.7	0.7	0.41	0.40	0.21*	0.22	0.18*	0.30	–0.64	–					
9. Pleasure	5.3	0.8	0.56	0.49	0.42	0.40	0.34	0.26	–0.58	0.61	–				
10. Enthusiasm	5.0	0.9	0.48	0.49	0.43	0.60	0.36	0.22	–0.43	0.34	0.72	–			
11. Vigor	4.9	0.7	0.34	0.36	0.31	0.34	0.24	0.19*	–0.37	0.32	0.49	0.67	–		
12. Placidity	5.1	0.8	0.49	0.45	0.32	0.24	0.32	0.20*	–0.49	0.54	0.60	0.46	0.37	–	
13. Overall affective well being	4.8	0.6	0.57	0.57	0.43	0.48	0.36	0.27	–0.61	0.67	0.88	0.82	0.74	0.73	–
14. Self-reported performance	5.2	0.8	0.26	0.28	0.33	0.41	0.16**	0.18*	–0.25	0.24	0.37	0.53	0.47	0.32	0.50

Correlations are significant for $p < 0.001$, except when they are signaled (* $p < 0.01$; ** $p < 0.05$).

and belongingness needs) and self-determination (e.g., people feel that they can be promoted by means of their merit and not through “personal favoritism”). Work–family conciliation meets the sense of self-determination (i.e., employees have a feeling of control over their lives). Opportunities for learning and personal development (comprising aspects related to empowerment, personal growth and putting imagination and creativity to work) also match the senses of impact, competence, enjoyment and meaning. Open and frank communication with the leader helps employees to develop the sense of impact (“my ideas are respected and my saying matters”). The trust and credibility of the leaders meet the senses of impact and competency because employees may feel free to propose unconventional ideas for solving problems and seize opportunities, and to introduce

conflicting opinions without fear of reprisal (Prati et al., 2002). Trust and credibility of the leaders also likely foster leader–member authentic relationships, thus promoting hope and optimism in employees and, thus, their senses of meaning, impact and enjoyment (Avolio, Gardner, Walumbwa, Luthans, and May, 2004; Gardner, Avolio, Luthans, May, and Walumbwa, 2005; Ilies, Morgeson, and Nahrgang, 2005). Open and frank communication with the leader, authenticity in the relationship and hope can also engender the sense of purpose (“I can participate in designing the future of my organization and I trust in the mission/vision my leaders recognize”).

Individuals responded on a six-point scale ranging between “the statement is completely false” (1) and “[it] is completely true” (6). Confirmatory factor analysis shows that the six-factor

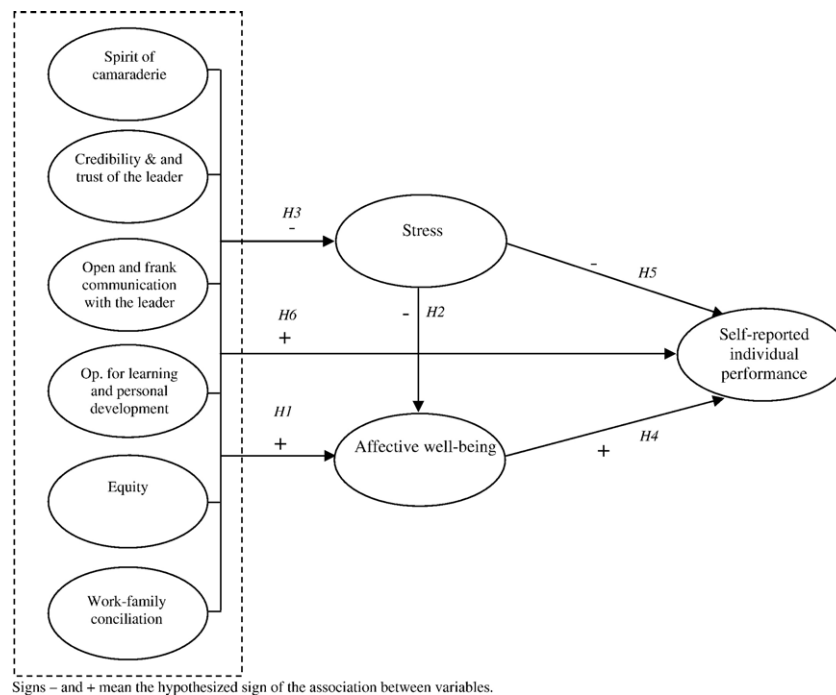


Fig. 1. Hypothesized model.

Table 4
Results of hypotheses (structural equation modeling)

Paths	Standardized path coefficients	Hypotheses	Hypotheses supported (yes/no)
Spirit of camaraderie → affective well being	0.23*	1a	Yes
Spirit of camaraderie → stress	−0.16	3a	No
Spirit of camaraderie → self-reported individual performance	−0.11	6a	No
Trust and credibility of the leader → affective well being	0.11	1b	No
Trust and credibility of the leader → stress	−0.45*	3b	Yes
Trust and credibility of the leader → self-reported individual performance	−0.17	6b	No
Open and frank com. with the leader → affective well being	0.03	1c	No
Open and frank com. with the leader → stress	0.10	3c	No
Open and frank com. with the leader → self-reported individual performance	0.19*	6c	Yes
Opp. for learning and pers. development → affective well being	0.19*	1d	Yes
Opp. for learning and pers. development → stress	0.13	3d	No
Opp. for learning and pers. development → self-reported individual performance	0.26*	6d	Yes
Fairness → affective well being	−0.00	1e	No
Fairness → stress	0.04	3e	No
Fairness → self-reported individual performance	−0.07	6e	No
Work–family conciliation → affective well being	−0.08	1f	No
Work–family conciliation → stress	−0.20*	3f	Yes
Work–family conciliation → self-reported individual performance	0.07	6f	No
Stress → affective well being	−0.43*	2	Yes
Stress → self-reported individual performance	0.08	5	No
Affective well being → self-reported individual performance	0.49*	4	Yes

The model is saturated. The fit is perfect.

*The path coefficient is significant for, at least, $p < 0.05$.

model fits the data well (Table 2). Only one lambda is lower than 0.50 (0.49), and alphas exceed or are very close to 0.70.

5.4. Independence between cases

An ANOVA was run with the organization as independent variable to test the independence between cases. For all the latent variables no significant differences between organizations are found (i.e., no F -values were significant; $p < 0.05$). This finding is not surprising considering that in most cases, individuals from the same organization came from different departments and sometimes

from different locations. Furthermore, as previously suggested, variation in perceptions is likely to result from differences among people, from differences in situations experienced by each employee (for example, regarding his/her specific exchange with supervisor), from the employee–situation interaction, and from perceptual biases.

6. Results

Table 3 presents the means, standard deviations, and correlations. Considering the six-point scale, the scores of

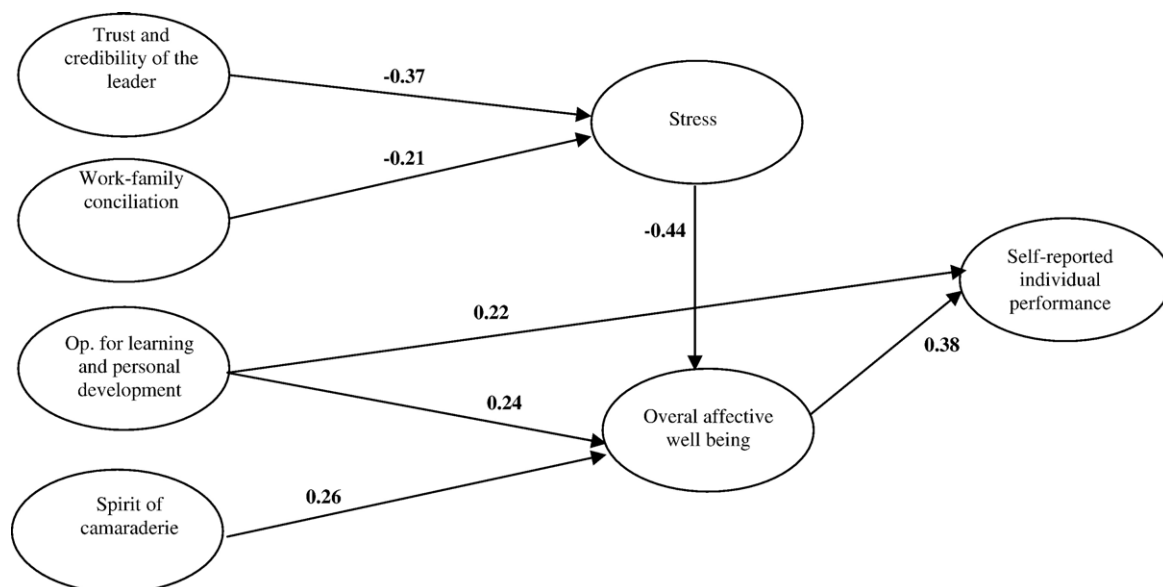


Fig. 2. Revised model.

Table 5
Hierarchical regression analyses: PWB and self-reported performance as dependent variables

	Comfort		Pleasure		Enthusiasm		Vigor		Placidity		Stress		Self-reported individual performance	
	1st step	2nd step	1st step	2nd step	1st step	2nd step	1st step	2nd step	1st step	2nd step	1st step	2nd step	1st step	2nd step
Spirit of camaraderie		0.23**		0.30***		0.12		0.07		0.26**		0.07	−0.02	−0.10
Trust and credibility of the leader		0.15		−0.00		−0.00		0.02		0.22*		−0.21*	−0.10	−0.17
Open and frank com. with the leader		−0.15		0.12		0.01		0.08		0.03		0.01	0.21*	0.24*
Opportunities for learning and personal development		−0.08		0.1		0.51***		0.20*		−0.15		0.17*	0.38***	0.21*
Fairness		−0.01		−0.01		0.02		0.02		0.07		0.05	−0.10	−0.12
Work–family conciliation		0.00		−0.09		−0.09		−0.03		−0.17		−0.18**	0.05	0.10
Stress	−0.65***	−0.55***	−0.52***	−0.40***	−0.38***	−0.24***	−0.35***	−0.26**	−0.48***	−0.36***			−0.11	−0.11
Comfort											−0.52***	−0.43***		0.17
Pleasure											−0.02	−0.03		−0.12
Enthusiasm											−0.10	−0.12		0.38**
Vigor											−0.06	−0.04		0.20*
Placidity											−0.15***	−0.13*		0.19*
F	133.2***	22.3***	67.6***	19.4***	29.4***	20.4***	25.4***	7.0***	54.9***	13.9***	32.58***	17.8***	6.6***	8.8***
Adjusted R ²	0.42	0.45	0.26	0.42	0.14	0.43	0.12	0.19	0.22	0.33	0.47	0.50	0.21	0.34
Unique variance imputable to the authenticizotic psychological climates		0.03		0.16		0.29		0.07		0.11		0.03		
Unique variance imputable to the affective well being variables														0.13

* $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$.

psychological climates are moderate, the lowest referring to work–family balance. Considering the seven-point scale, affective well being is moderate (especially the comfort score), the level of stress is low, and employees tend to self-report moderate performance. Psychological climates inter-correlate positively, but the correlations concerning work–family balance are moderate. Psychological climates correlate negatively with stress and positively with affective well being. The dimensions of affective well-being inter-correlate positively. Stress correlates negatively with all dimensions of affective well being. Self-reported performance correlates positively with authenticzotic and affective well being variables, and negatively with stress.

The study tests the hypotheses (Fig. 1) through structural equation modeling using LISREL with the maximum likelihood estimation method. For avoiding an excessively complex model (17 variables and 53 paths), the overall affective well being score is used instead of all affective well being dimensions. Table 4 depicts the standardized path coefficients for each hypothesis. The findings support hypotheses 1a, 1d, 2, 3b, 3f, 4, 6c and 6d.

The analysis includes examining a revised model comprising only the significant paths. The path between open and frank communication with the leader and self-reported individual performance is non-significant. The resulting model (Fig. 2) fits the data well: the root mean square error or approximation is 0.06, the goodness of fit index is 0.98, the adjusted goodness of fit index is 0.93, the comparative fit index is 0.99, the incremental fit index is 0.99 and the relative fit index is 0.93. The main results are as follows: (a) perceptions of spirit of camaraderie and opportunities for learning and personal development predict overall affective well being and, through the mediation of this variable, they also predict self-reported performance; (b) the perceptions of opportunities for learning and personal development predict directly self-reported performance; (c) perceptions of trust and credibility of the leader, as well as the perceptions of work–family conciliation, predict stress directly and, through this variable, overall affective well being. The findings show that psychological

climates relate with both variables of psychological well being and, through these variables or other ones not studied here, they also relate with the self-reported performance.

Hierarchical regression analyses explore the role played by each affective well being dimension (Table 5): (a) stress continues to explain all the affective well being dimensions after entering all the psychological climates; (b) the perceptions of spirit of camaraderie explain unique variance of affective well being, mainly the comfort, pleasure and placidity dimensions, and the perceptions of opportunities for learning and personal development explain unique variance of affective well being, mainly the enthusiasm and vigor dimensions; (c) the perceptions of trust and credibility of the leader and of opportunities for learning and personal development continue to predict stress after entering affective well being dimensions; (d) affective well being explains unique variance of self-reported individual performance, the main predictors being enthusiasm, vigor and placidity.

In predicting stress, the positive beta regarding opportunities for learning and personal development is surprising, considering that both variables inter-correlate negatively. Adding to the regression the product of those opportunities by work–family conciliation, the predictive value for stress increases 1.1% and the beta of the product of this interaction is significant. Fig. 3 depicts the pattern of the interaction: (a) stress increases when people feel that they cannot take advantage of opportunities for learning and personal development due to some work–family conflict; (b) stress decreases when people feel that those opportunities are aligned with good conditions to balance work and family roles.

7. Discussion and conclusion

7.1. Making sense of the main findings

The findings corroborate the notion that the construct of psychological climate is “alive and well” (Parker et al., 2003, p. 408). Authenticzotic psychological climates may be relevant for stress and affective well being. These climates may also matter for individual performance, but the “fragile” nature of the performance measurement recommends modesty and caution in interpreting findings. Next, the discussion focuses on the association with PWB and then with individual performance.

First, employees show greater stress when they perceive that their leaders lack trustworthiness and credibility, and that conditions to balance work and family roles are absent. Lack of trust in the leader may be a source of psychological insecurity. The perceptions of lack of work–family balance may also lead to role conflict and induce feelings of poor organizational support, thus increasing stress. And the opportunities of learning and personal development can foster stress if employees sense a lack of conditions to balance family and work roles.

Second, employees who perceive a positive spirit of camaraderie report greater comfort, pleasure and placidity. They meet social needs, obtain social support for dealing with work difficulties, challenges and opportunities, experience fewer relationship conflicts and feel intrinsically motivated (Kasser and Ryan, 1996) for being respected as human beings and not just

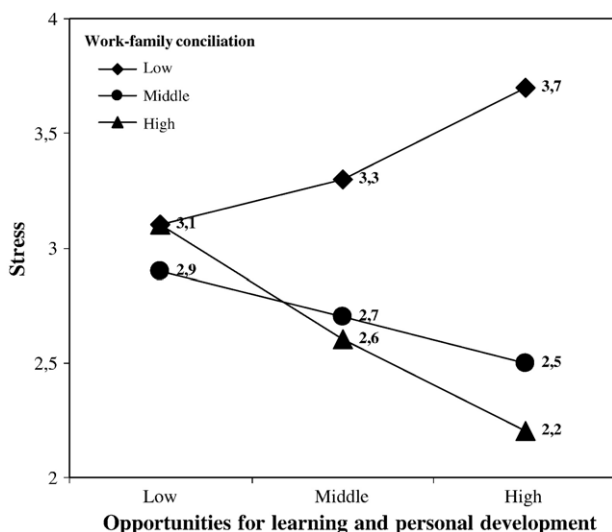


Fig. 3. How work–family conciliation and opportunities for learning and personal development interact to predict stress.

as “resources”. Employees perceiving better opportunities for learning and personal development develop more enthusiasm and vigor at work. They perceive their jobs as more intrinsically motivating and rewarding, feel more empowered, feel they can put creativity into the job, and perceive that their senses of impact, competence, enjoyment and meaning are being fulfilled.

Psychological climates predict self-reported individual performance, either directly or through PWB variables. Indeed, one direct significant path between the perceptions of good opportunities for learning and personal development and self-reported individual performance is found. The perceptions of these good opportunities induce employees to get meaning from performing the job, to feel that they can develop their competencies and to put their full potential into carrying out the job. But psychological climates also seem relevant for individual performance because they predict affective well being, which in turn predicts performance. In fact, employees who experience greater versus lower affective well being (mainly enthusiasm, vigor and placidity) report higher performance. The enthusiastic, vigorous and “calm” employees express more commitment to work, apply their potential in performing the job, actively try to solve problems and take advantage of opportunities, and persevere calmly when facing obstacles (Frederickson, 2001; Wright and Cropanzano, 2004).

Contrary to the hypothesized, stress does not explain unique variance of self-reported performance, but lower versus higher stress associates with greater affective well being. This finding corroborates the literature suggesting that stress relates with low levels of job satisfaction and PWB (Daniels and Guppy, 1994; Deborah et al., 1993; Fairbrother and Warn, 2003). Thus, the effect of stress on individual performance may be indirect, and stress may be detrimental to performance due to the association with lower affective well being.

Equity does not show any predictive value for the independent variables. This is surprising, considering that justice has been said to impact several outcomes: work performance, stress, counter-productive behaviors, work satisfaction, organizational commitment, trust in the organization and its leaders, turnover intentions and emotions (Rego, Cunha, and Lopes, *in press*). One possible explanation is that the equity measure is one-dimensional and does not capture the richness of justice as a multidimensional construct (e.g., distributive; procedural; interactional).

The findings are consistent with findings in the literature (e.g., Baumeister and Leary, 1995; Brown and Leigh, 1996; Deborah et al., 1993; Martin et al., 2005; Parker et al., 2003) which suggests that employees experience higher PWB, are more committed to work and invest their entire selves in the job, when (a) they perceive that the work environment is positive, respectful, supportive, psychologically safe and meaningful and that (b) such environment provides conditions to satisfy their social, intimacy, security, emotional and learning/exploration needs.

7.2. Implications for management

Promoting a psychologically happy workforce is a valuable goal in itself, as the “positive organizational studies” and the “economics of happiness” (Graham, 2005) movements suggest.

However, the study also suggests that promoting a psychologically happy workforce may be a means for boosting individual performance if leadership and organizational practices encourage employees’ positive perceptions of features of the work environment. The performance measurement is open to criticism. However, considering both theoretical and empirical evidence that suggest that self-ratings associate with ratings from peers and supervisors and with objective measures of productivity (see previous section in the paper), the findings provide orientation for managers aiming to foster individual performance.

To build authentic psychological climates and healthy organizations, managers must care about how employees perceive the organization, paying attention to a number of aspects: (a) a respectful and trustful way of acting; (b) opportunities for employee learning and personal development; (c) the degree to which they treat employees as people in search of meaningful work; (d) the honesty and frankness they place in relating to subordinates; (e) strategies they develop to facilitate work–family balance; (f) ways they promote spirit of camaraderie and teamwork and (g) the fairness in their decisions involving promotions and rewards. If, as Haller and Hadler (2006) argue, the people’s subjective perception and evaluation is most significant for PWB, managers must act to influence employees’ perceptions, not only for changing the work environment. These are reasons to follow Parker et al. (2003, p. 406) recommendation: “psychological climate assessments should be part of interventions attempting to improve the quality of work life (...)”, to reduce employee turnover and to improve motivation and performance.

Such recommendation does not mean that merely managing perceptions is enough or recommendable. Leaders and organizations risk nourishing employees’ cynicism if they adopt a facade of authentic workplaces, a “swindler” or “hollow” healthy workplace (Greenberg, 1990) without evidence of genuinely positive and respectful actions and decisions. In the long run, the consequences of that “swindle” may be cynicism, which in turn declines the employee commitment and performance, and increases retaliation behaviors (Andersson and Bateman, 1997; Dean, Brandes, and Dharwadkar, 1998). “Managing” perceptions is necessary but not enough for promoting a healthy workplace. Acting in an authentic way is necessary but not enough to create a healthy workplace. Managers and organizations must be cognizant of employees’ perceptions and willing to make adjustments accordingly, but they should not focus solely on employees’ perceptions — rather they must actively operate upon the sources of organizational health, the more effective way to foster positive perceptions.

7.3. Limitations and future research

The dependent and independent variables were collected simultaneously from the same source. This makes the study vulnerable to common method variance. Future studies could collect data for dependent and independent variables at separate moments. This would reduce the respondents’ tendency to

search for similarities in the questions and to maintain consistency in the answers. Another way to minimize these risks is to use a multiple source method, with some individuals reporting their perceptions of organizational features, and others expressing their levels of stress and well being, and supervisors describing their performance. Objective measures of performance, as well as subjective measures from peers, may also be considered when possible. Although theoretical and empirical research suggests that self-assessments are of great value (Fox and Dinur, 1988; Levine, 1980; Van der Heijden and Nijhof, 2004), one needs to read the findings with caution.

The precautions the methods section mentions to reduce method biases seem to be fair and reasonable remedies (Podsakoff, MacKenzie, Lee, and Podsakoff, 2003). As a complementary “statistical remedy”, the study includes performing Harman’s single-factor analysis. If a significant amount of common method bias exists in the data, a factor analysis (unrotated solution) of all the variables in the model would give rise to a single factor accounting for most variance. An eigenvalue-greater-than-one criterion reveals eleven factors, the first of which explains 27% of the variance. This result suggests that the data are not subject to common method bias.

Another criticism is that the study does not express the causal links between dependent and independent variables, and other causal links are plausible as well. For example: (a) the employees’ affective states may influence their perceptions of the organizational features (Brief and Weiss, 2002), rather than the other way around; (b) stress can influence emotions, but positive emotions can also help people to be more proactive and resilient, and less prone to stress symptoms (Folkman, 1997; Ong, Bergeman, Bisconti, and Wallace, 2006; Tugade and Fredrickson, 2004). Post-hoc rationalizations can also explain some findings (e.g., people who feel good over recent months in the organization may cognitively “reciprocate”, attributing positive features to the organization, independently of the “real” characteristics). Future longitudinal or quasi-experimental studies must be conducted for specifying what the more robust causal links are. All respondents are university graduates. Future studies may gather a more diverse and larger sample.

7.4. Concluding remarks

Despite the above criticisms, the study advances two important points: (1) organizations and leaders can promote a psychologically healthy workforce if they improve (in a genuine and sustainable way) the perceptions of their employees regarding the dimensions discussed in this paper; (2) this may produce good results, considering that employees experiencing higher PWB may also be more productive.

Brief and Weiss (2002, p. 299) state that “we know less than we should about features of work environments that are likely to produce particular (positive and negative) moods and emotions among those who spend perhaps the majority of their working hours in them, five or more days a week”. As Wright and Cropanzano (2004, p. 338) point out, “It is reasonable and highly practical for both business executives and management scholars to understand that happiness is a valuable tool for

maximizing both personal betterment and employee job performance”. The present article offers empirical evidence for both arguments.

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